CallWise

Whom should a bank call? Trees, forests and a famous leak

EC-417 Introduction to Machine Intelligence · Term Project P05

Roll Nos. 623169 · 623195 · NIT Andhra Pradesh · November 2026

0.44

test PR-AUC without the leaky feature (call length)

3.8× the no-skill line of 0.117

Always saying “no” is 88% accurate and useless

The task: rank clients before the call

Estimate each client’s chance of subscribing to a term deposit, then call the best-ranked clients first.

Primary metric: PR-AUC

Rewards putting the rare “yes” clients at the top of the list. Its no-skill value is the positive rate, 0.117.

Also reported

ROC-AUC (optimistic on imbalanced data), recall at a threshold chosen on validation, lift and profit.

88.3%

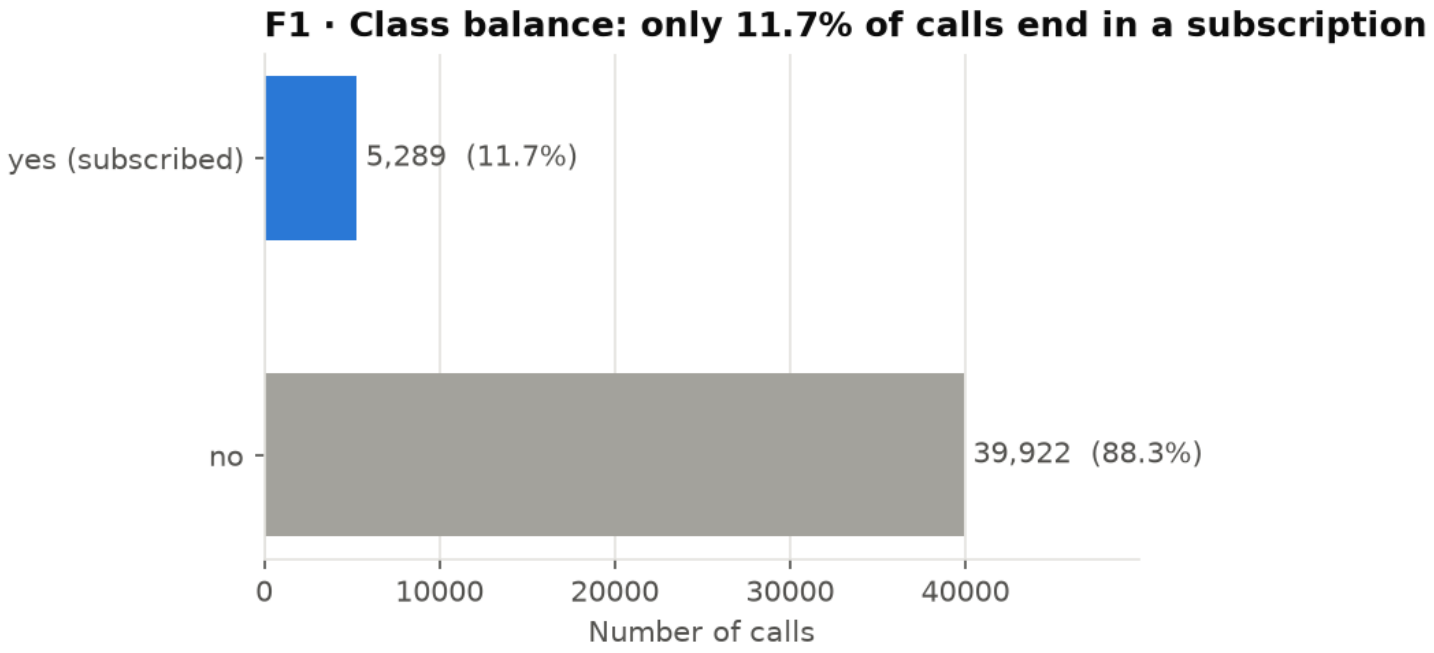
accuracy of always predicting “no”

0

subscribers that model finds

So we rank by probability and score the ranking.

45,211 calls, and only 11.7% end in a yes



Source: UCI Bank Marketing, bank-full.csv (Moro, Cortez and Rita, 2014), CC BY 4.0. “unknown” is kept as its own category; pdays = −1 means never contacted, so a previously_contacted flag is added.

45,211

phone calls, May 2008 – Nov 2010

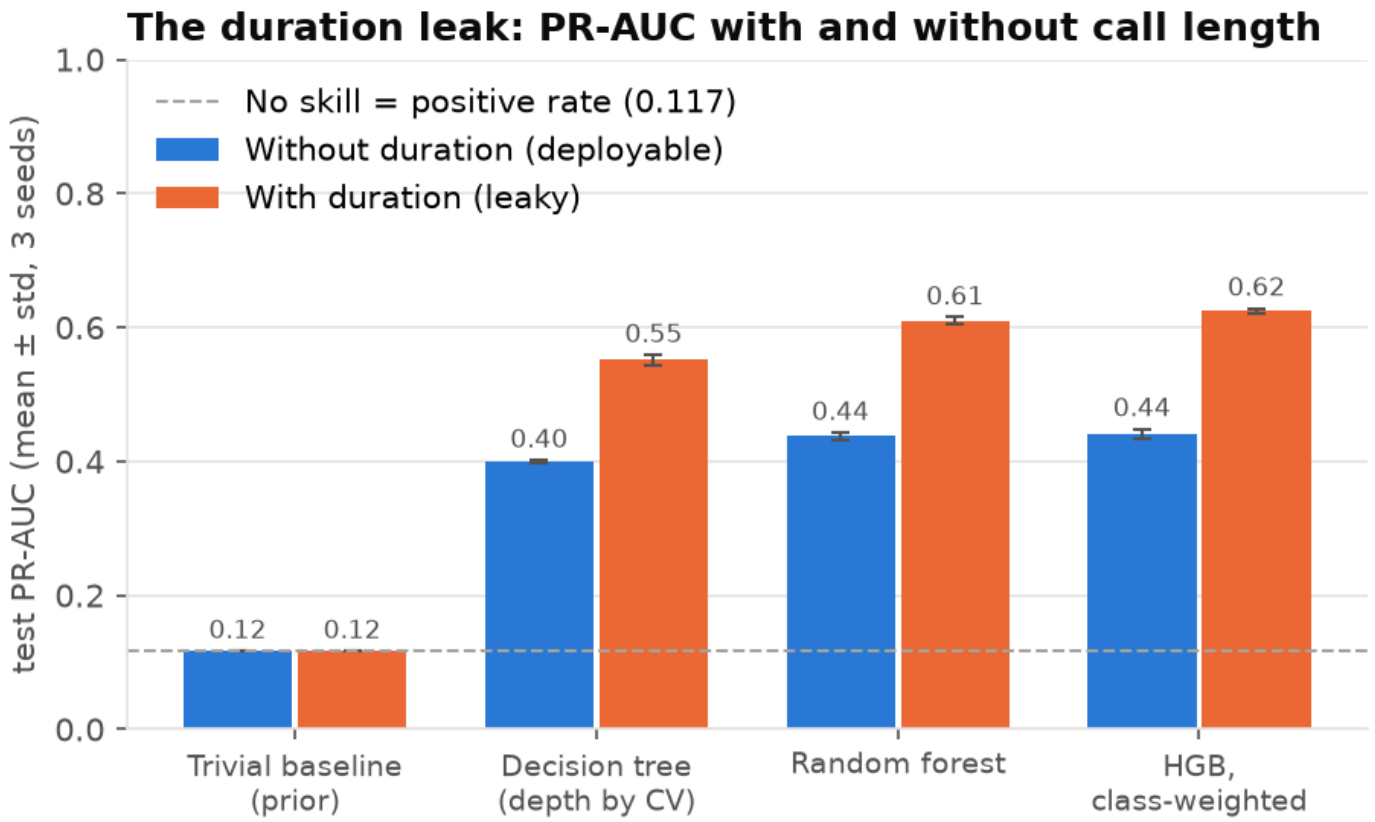
16

features about the client, the contact and past campaigns

5,289

calls ended in a subscription (11.7%)

Call length leaks the answer: +0.18 PR-AUC



Boosting, test PR-AUC

0.625 → 0.441

with → without duration

- duration is the length of the call: known only after it ends, and a 0-second call is always a “no”.
- Every model is trained twice: FULL (reference) and DEPLOYABLE (no duration).
- **All headline numbers use DEPLOYABLE.**

Split first: the test set is locked away



Seeds 42, 43, 44 reshuffle train ↔ validation

Same rows for every model and seed

1

Split first

Stratified 70/15/15 before anything is fitted.

2

Fit on train only

One-hot encoding sits inside the Pipeline; boosting reads categories natively.

3

Tune on validation

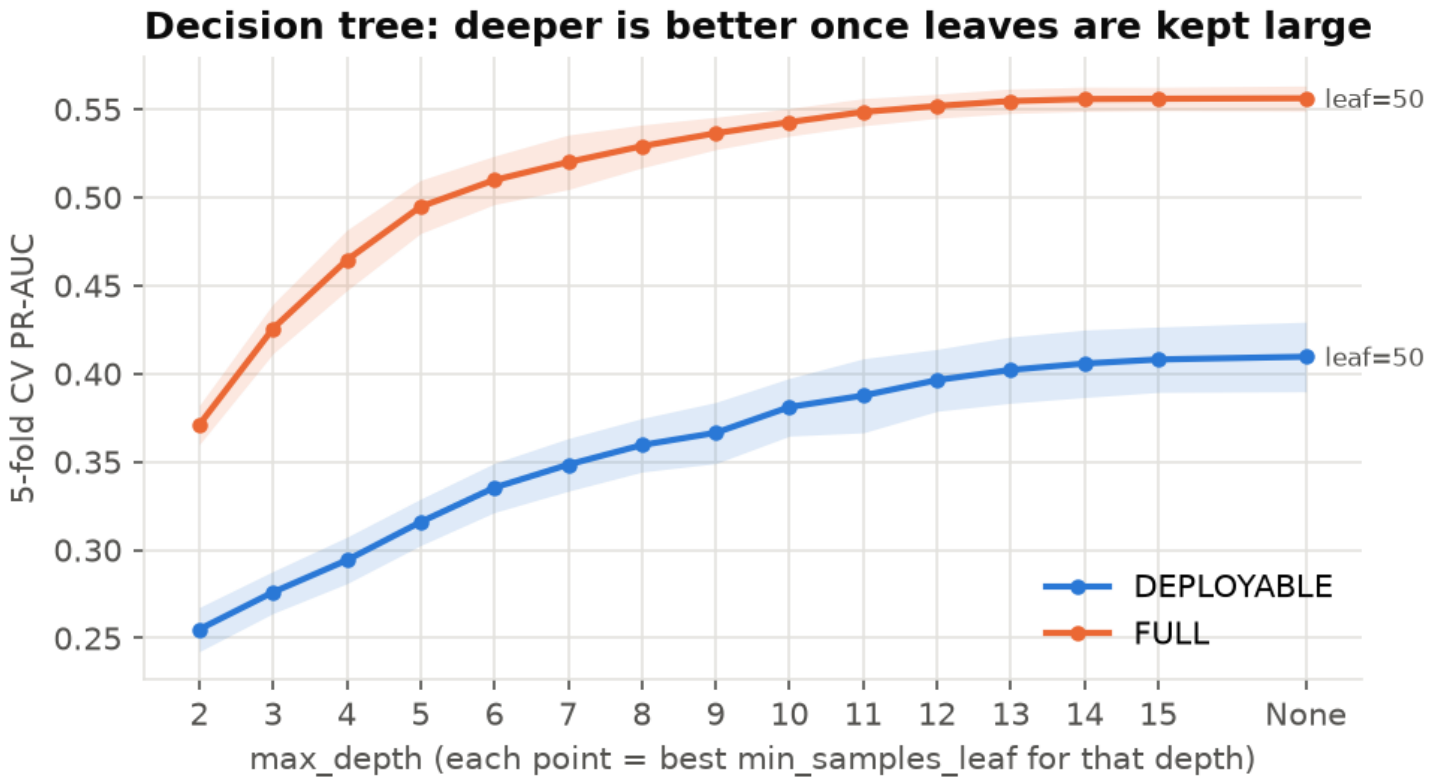
Tree depth by 5-fold CV; forest and boosting settings and the threshold on validation.

4

Score test once

Everything frozen first; report the mean ± std over 3 seeds.

A single decision tree reaches 3.4× no skill



0.117

Trivial baseline: predicts the 11.7% rate for everyone

0.399 ± 0.003

Decision tree, test PR-AUC (ROC-AUC 0.755)

5-fold CV chose unlimited depth with large leaves (50–100 clients each): leaf size, not depth, limits overfitting.

Boosting ties the forest; class weights don’t help

Random forest

main model

300 deep trees on bootstrap samples, averaged. 18 settings compared on validation.

0.438 ± 0.005

test PR-AUC

Weighted boosting

the improvement

Shallow trees added one at a time, each fixing current errors. Each “yes” weighs 7.5× a “no”.

0.441 ± 0.007

test PR-AUC

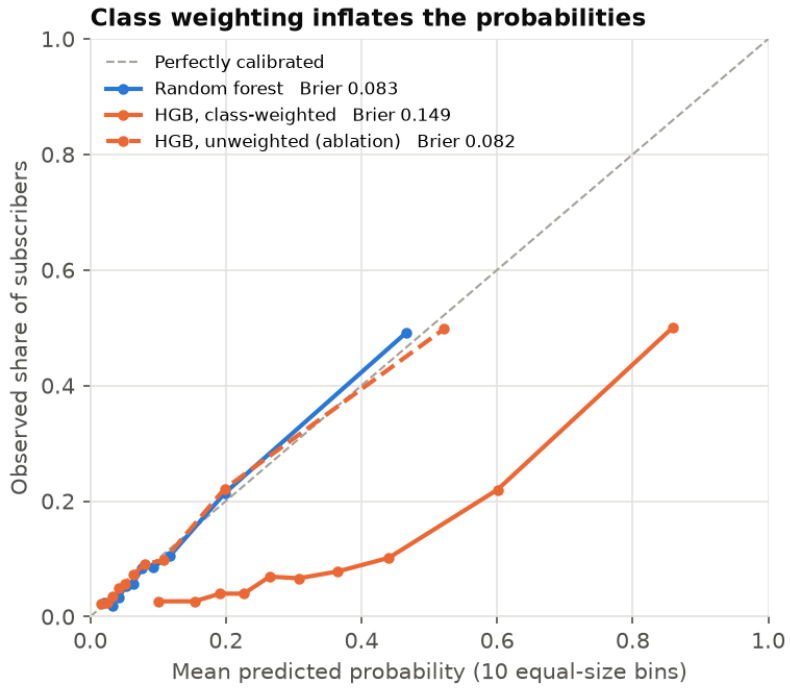
Ablation: the same boosting model with weighting switched off

0.446 ± 0.003

test PR-AUC (weighted: 0.441)

0.082 vs 0.149

Brier, unweighted vs weighted (lower = better)



Test set, seed 42. Brier: mean over 3 seeds (lower is better).

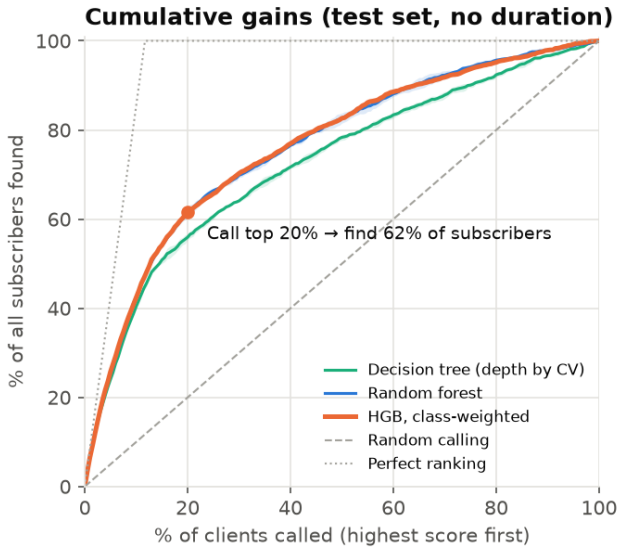
Weighting pushes scores up without changing their order: a weighted score of 0.6 means about a 22% chance.

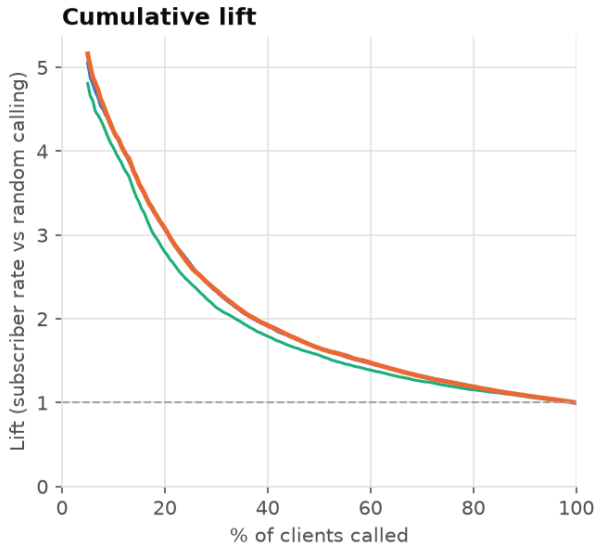
Without the leak, the ensembles lead at 0.44

Model	Features	PR-AUC	ROC-AUC	Recall @ t*	Train time
Trivial baseline (prior)	DEPLOYABLE	0.117 ± 0.000	0.500 ± 0.000	–	–
Decision tree (depth by CV)	DEPLOYABLE	0.399 ± 0.003	0.755 ± 0.001	0.475 ± 0.012	0.3 s
Random forest	DEPLOYABLE	0.438 ± 0.005	0.796 ± 0.002	0.521 ± 0.011	3.8 s
+ Boosting, class-weighted	**DEPLOYABLE**	**0.441 ± 0.007**	**0.797 ± 0.001**	**0.520 ± 0.013**	**1.7 s**
Boosting, unweighted (ablation)	DEPLOYABLE	0.446 ± 0.003	0.801 ± 0.002	0.497 ± 0.020	1.2 s
Decision tree	*FULL (leaky)*	*0.551 ± 0.007*	*0.901 ± 0.005*	*0.683 ± 0.054*	*0.4 s*
Random forest	*FULL (leaky)*	*0.609 ± 0.006*	*0.927 ± 0.000*	*0.715 ± 0.036*	*1.8 s*
Boosting, class-weighted	*FULL (leaky)*	*0.625 ± 0.004*	*0.931 ± 0.001*	*0.683 ± 0.024*	*1.9 s*

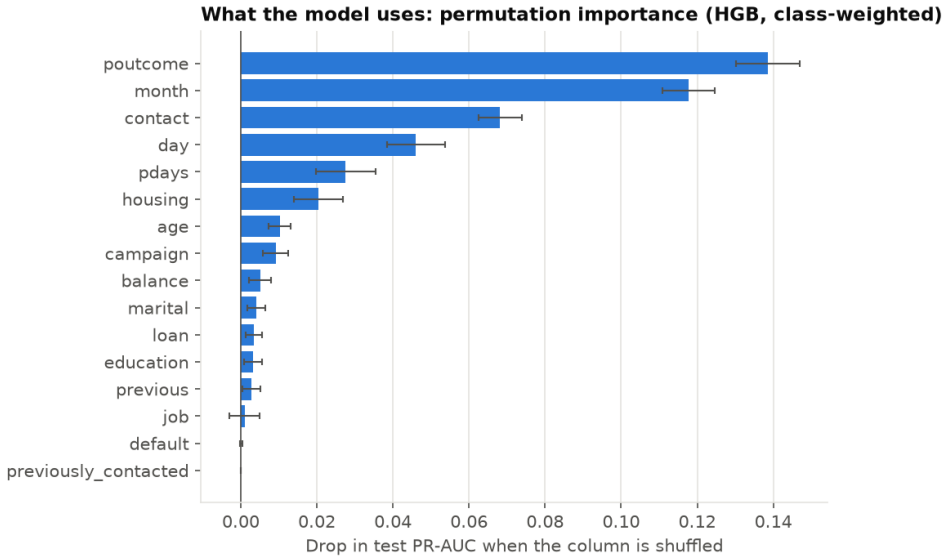
Test set: 6,782 clients (793 subscribers), scored once. Mean ± std over seeds 42, 43, 44. Threshold t* = best F1 on validation. Grey rows include the leaky duration feature and are for reference only.

Call the top 20% and find 62% of subscribers





Lines: mean over seeds 42, 43, 44; band: ±1 std.



Mean ± std over 10 shuffles x 3 seeds, test set, no duration.

42%

of subscribers found by calling only the top 10%

62%

found in the top 20%: 3.1× better than random calling

17.6k

profit vs 5.7k for calling everyone (cost 5, reward 50)

poutcome

and month matter most; demographics add little

What this model cannot tell us yet

Limitations

- Random, not time-based, split: no evidence yet about a future campaign
- One Portuguese bank, 2008–2010
- Weighted scores rank well but are not probabilities

Ethics

- The model calls 68% of clients aged 60+ and 51% of students, but about 9% of those aged 40–59
- Cap contacts per client and audit who is targeted

Next steps

- Train on earlier months, test on later ones
- Calibrate (isotonic), then set the profit threshold at c/r
- Better features for first-time contacts (82% of clients)

Code, results and report: github.com/sahitya1903/ec417-miniproject-P05-623169-623195 (tag v1.0)